Lab 05: Build, Test, and Package Applications with IntelliJ IDEA and Docker

Name: Fatima Kalsoom

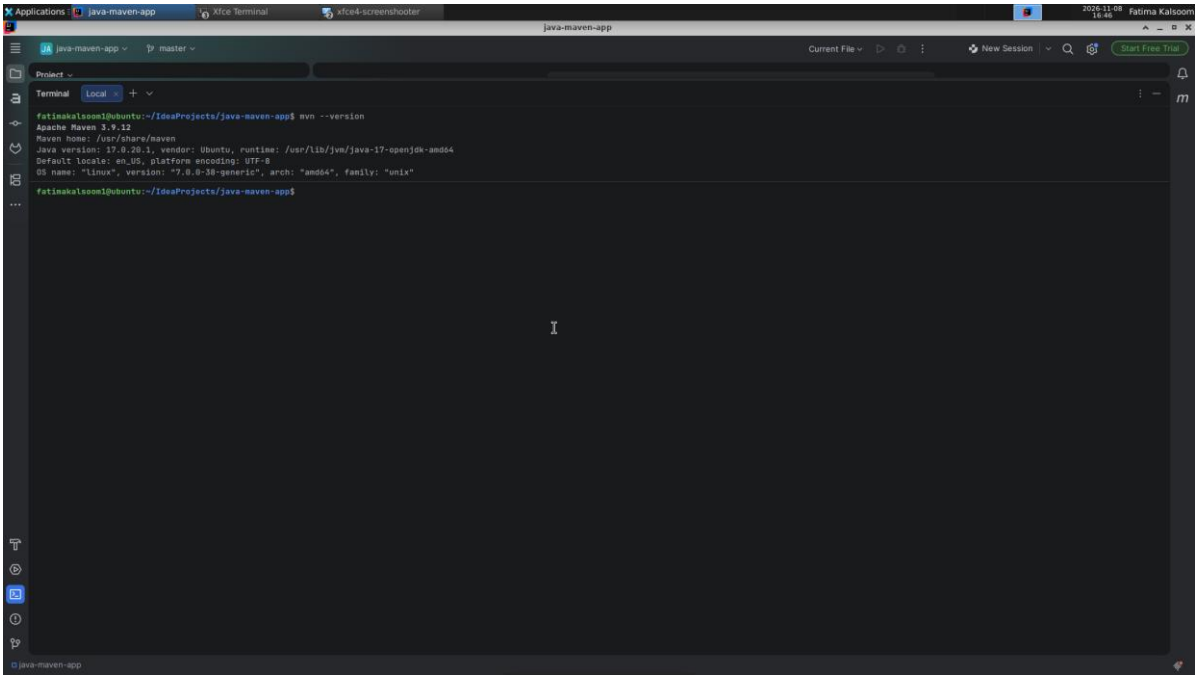
Registration Number: 2023-BSCS-32

Course Code: CS-649

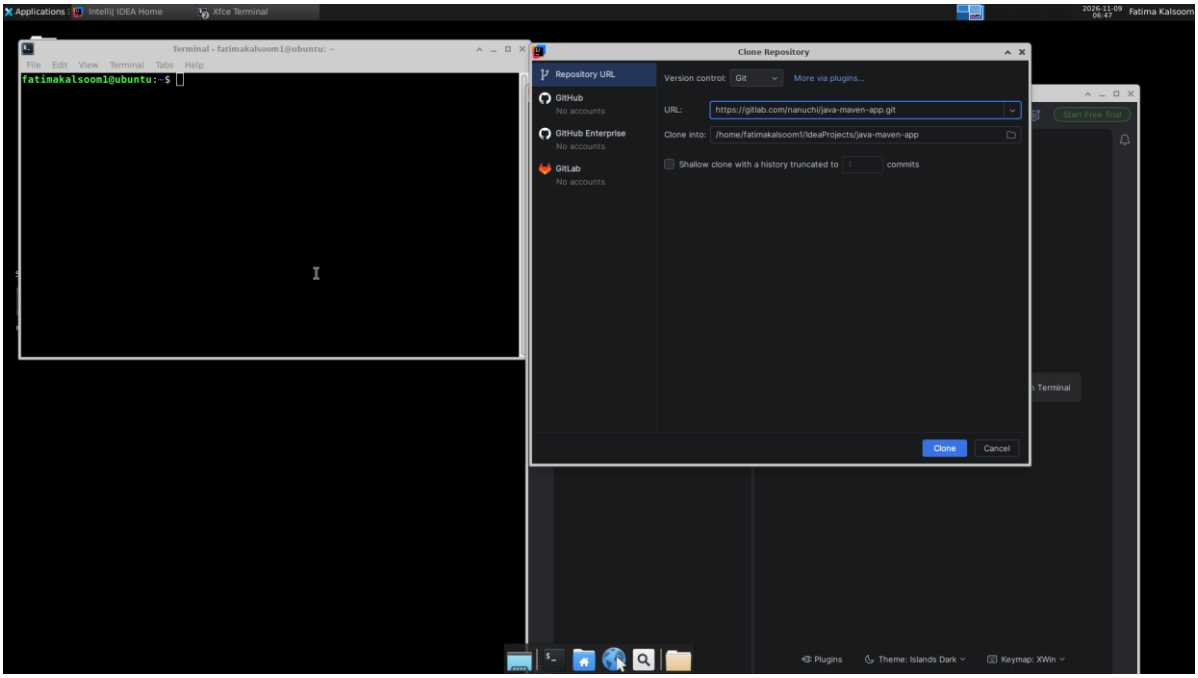
Section: A

GitHub Username: fatimakalsoom1

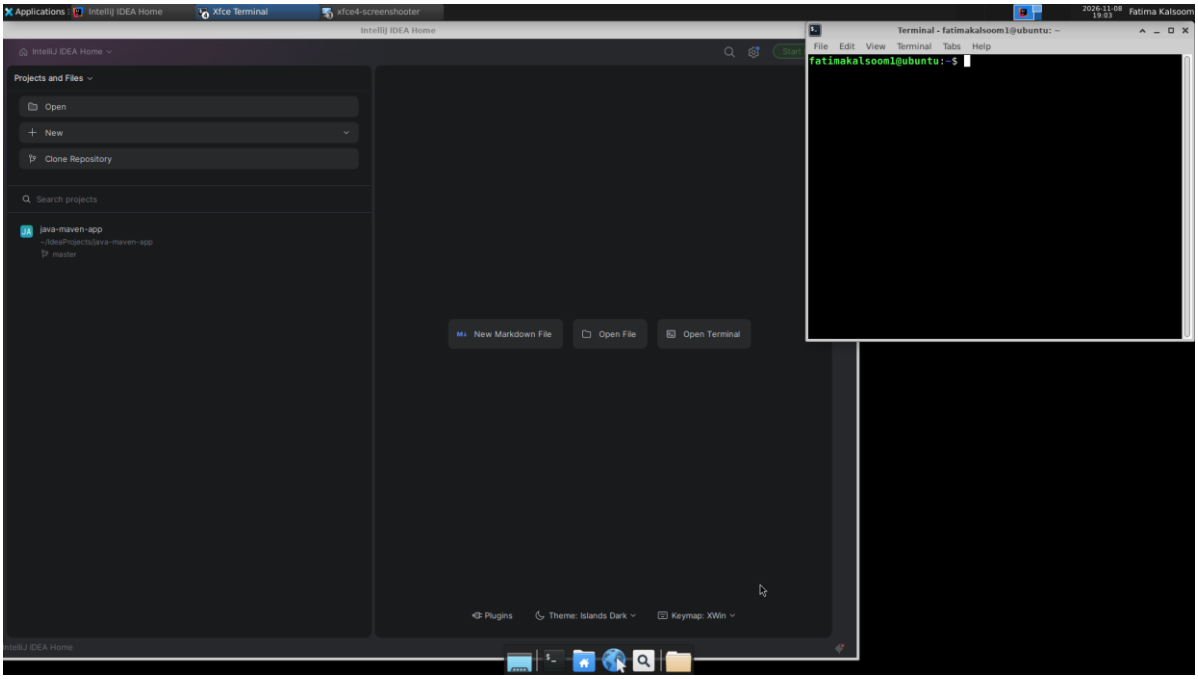
Task 1: Build a Java Application with Maven



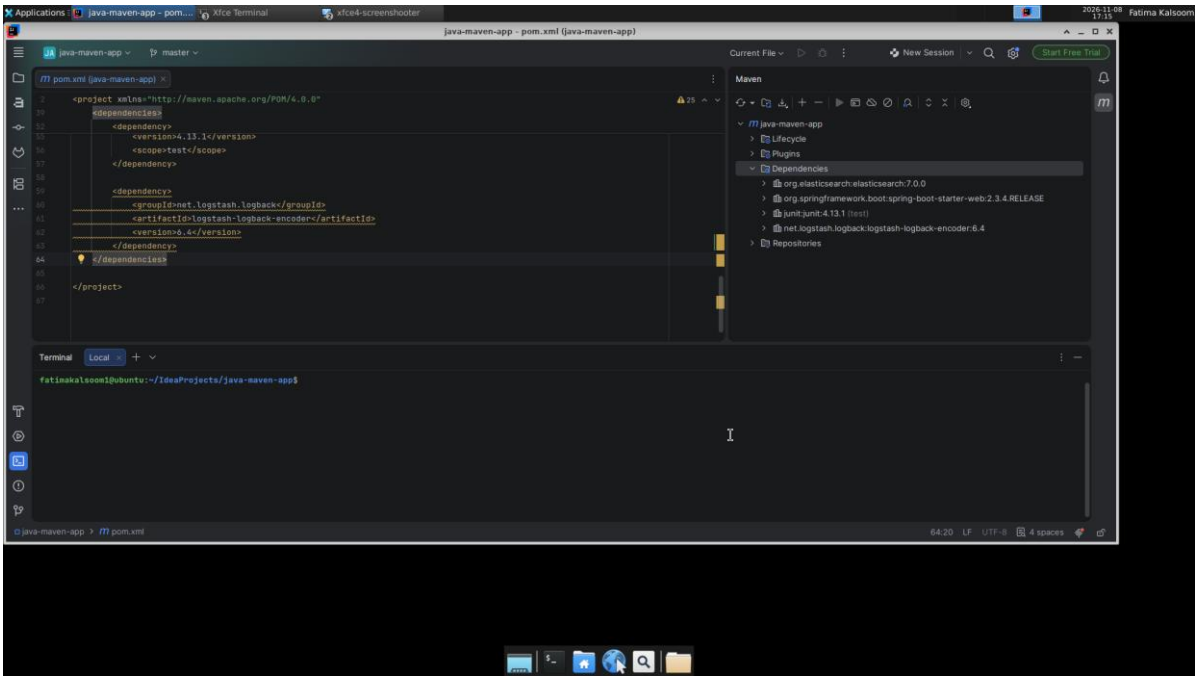
Verifying the Maven installation with mvn --version.



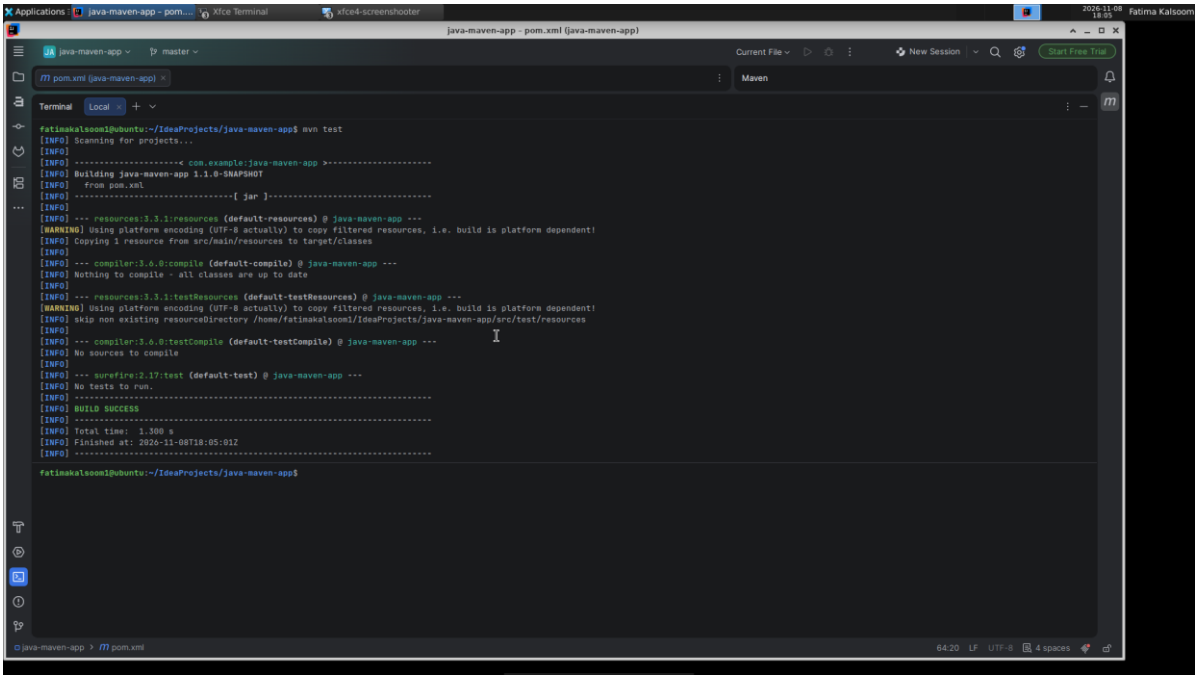
Cloning the java-maven-app repository in IntelliJ IDEA.



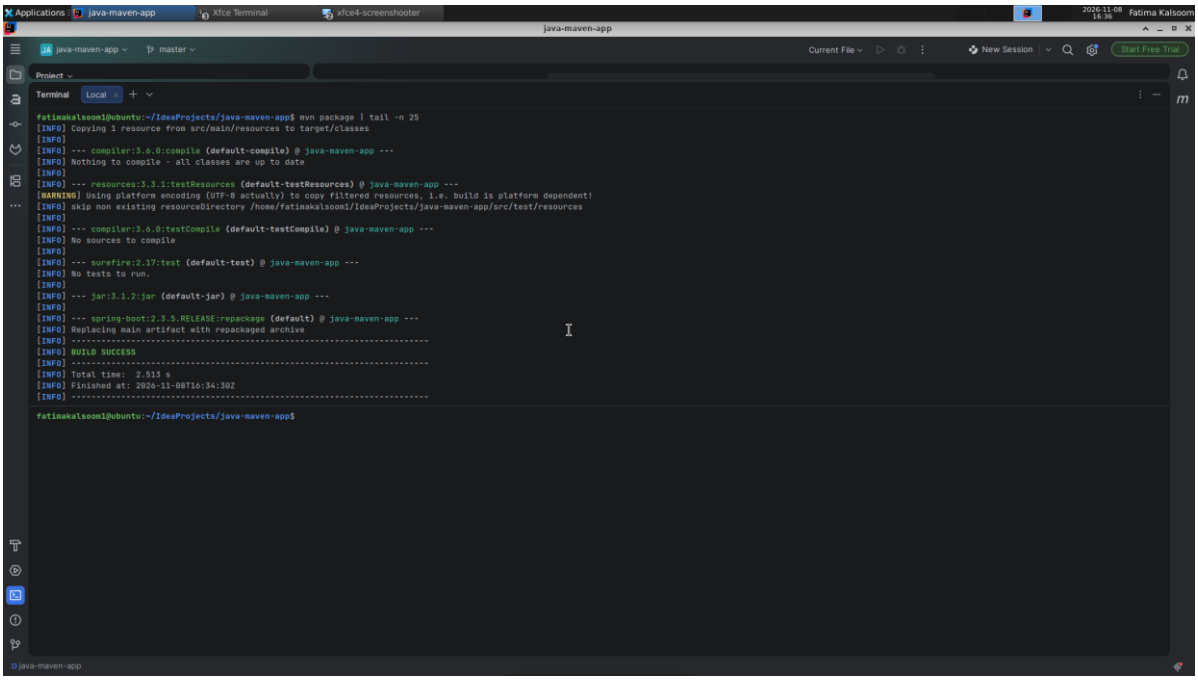
The java-maven-app project listed on the IntelliJ IDEA welcome screen.



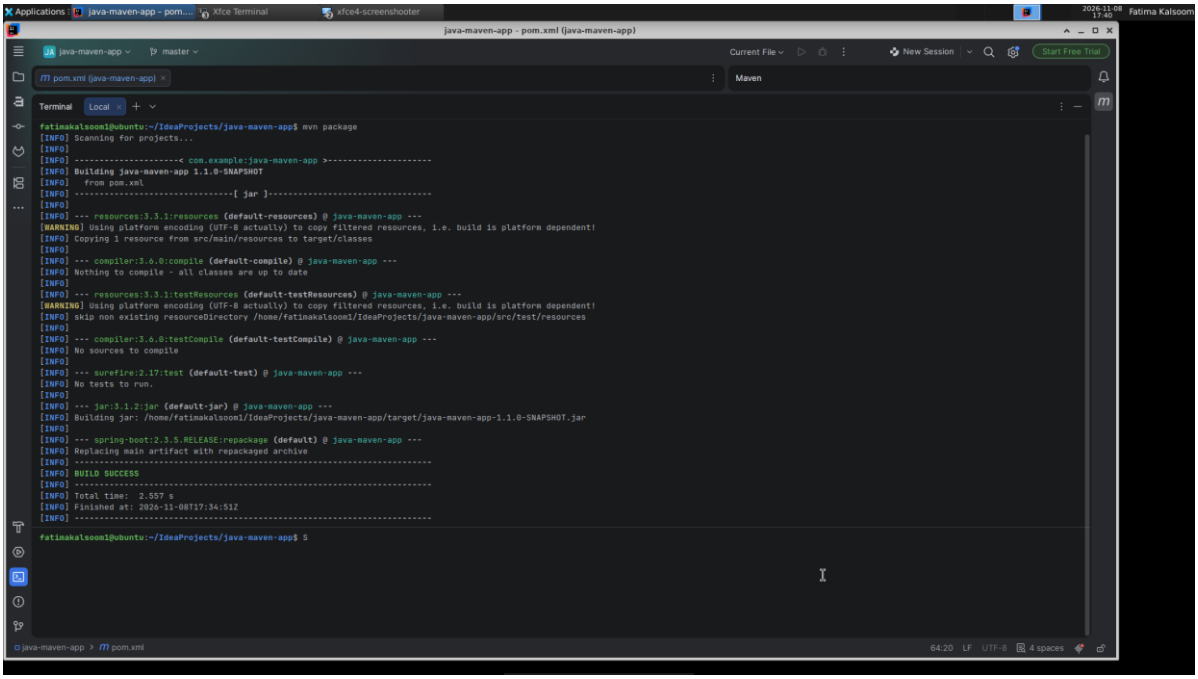
Elasticsearch dependency added to pom.xml and loaded in the Maven panel.



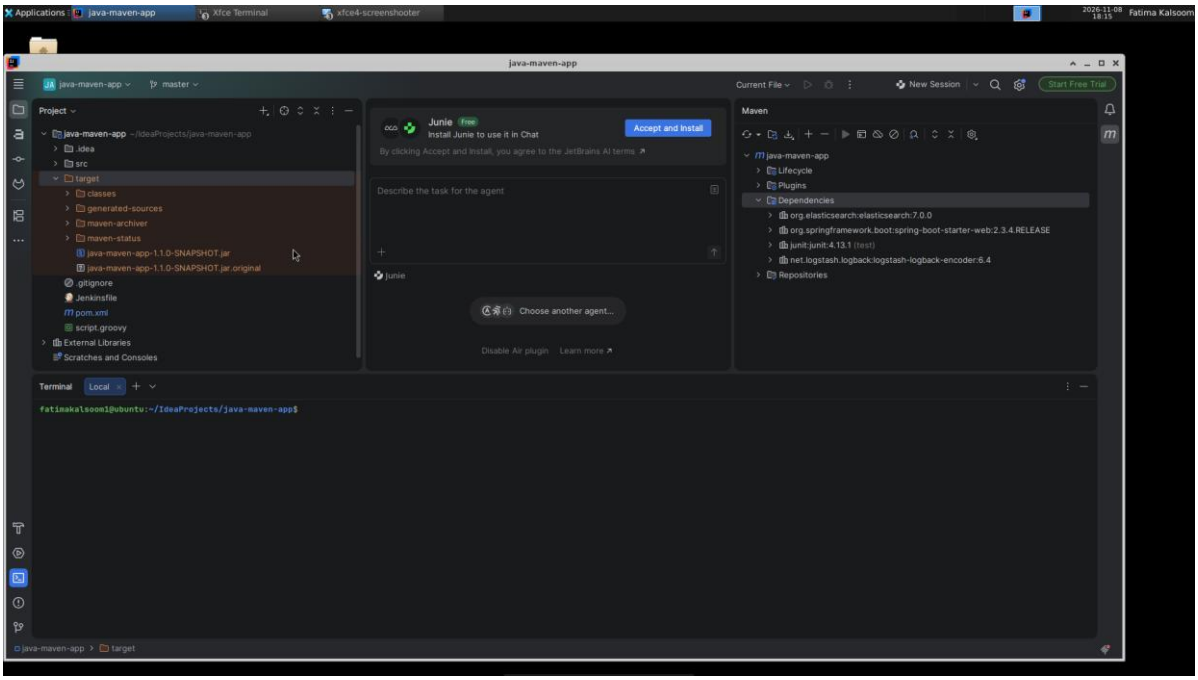
Running the tests with mvn test (BUILD SUCCESS).



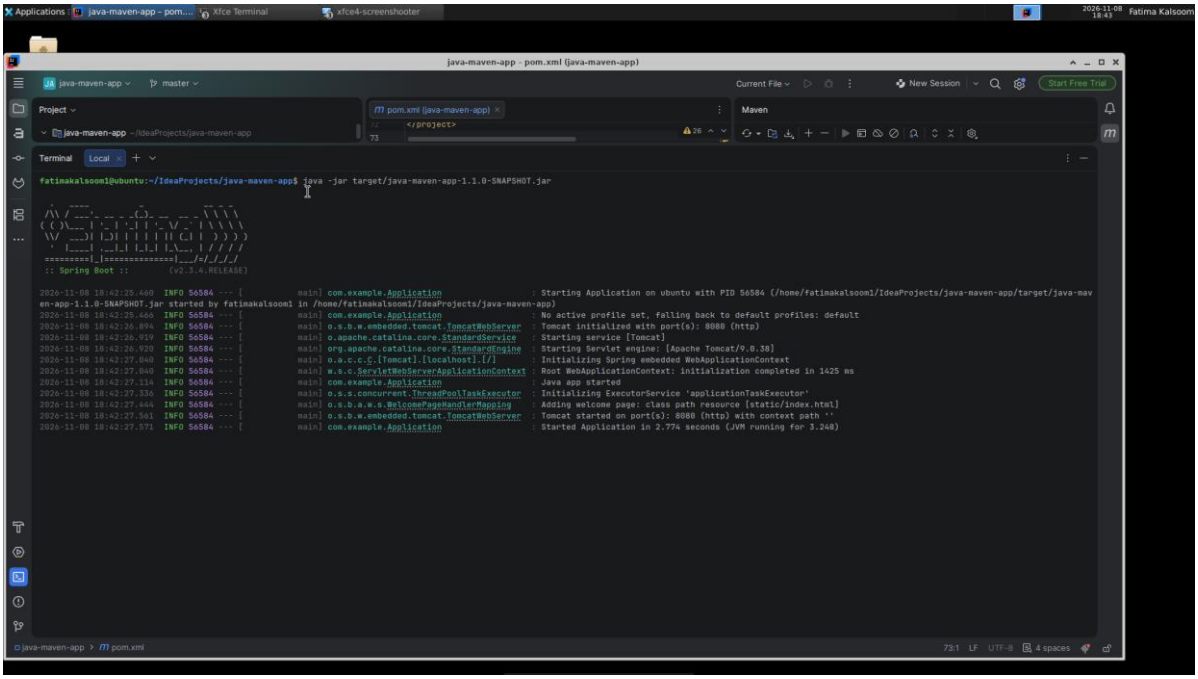
Packaging the application with mvn package (BUILD SUCCESS).



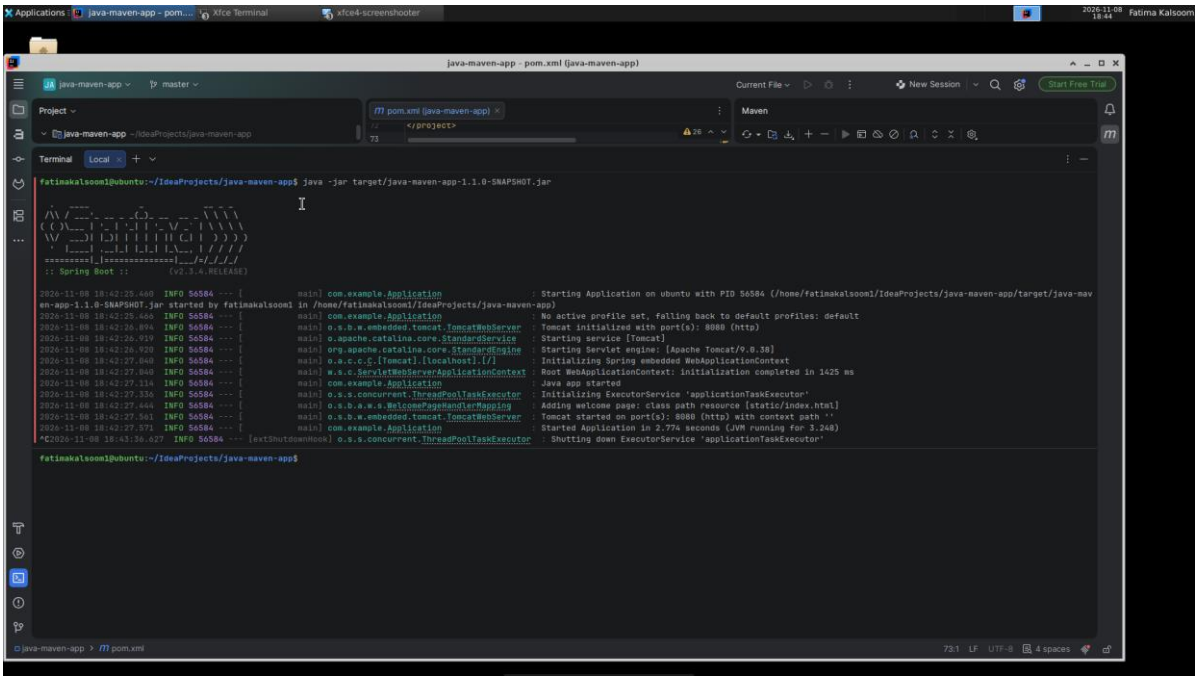
Re-packaging the application with mvn package after the changes.



The generated JAR file inside the target folder.

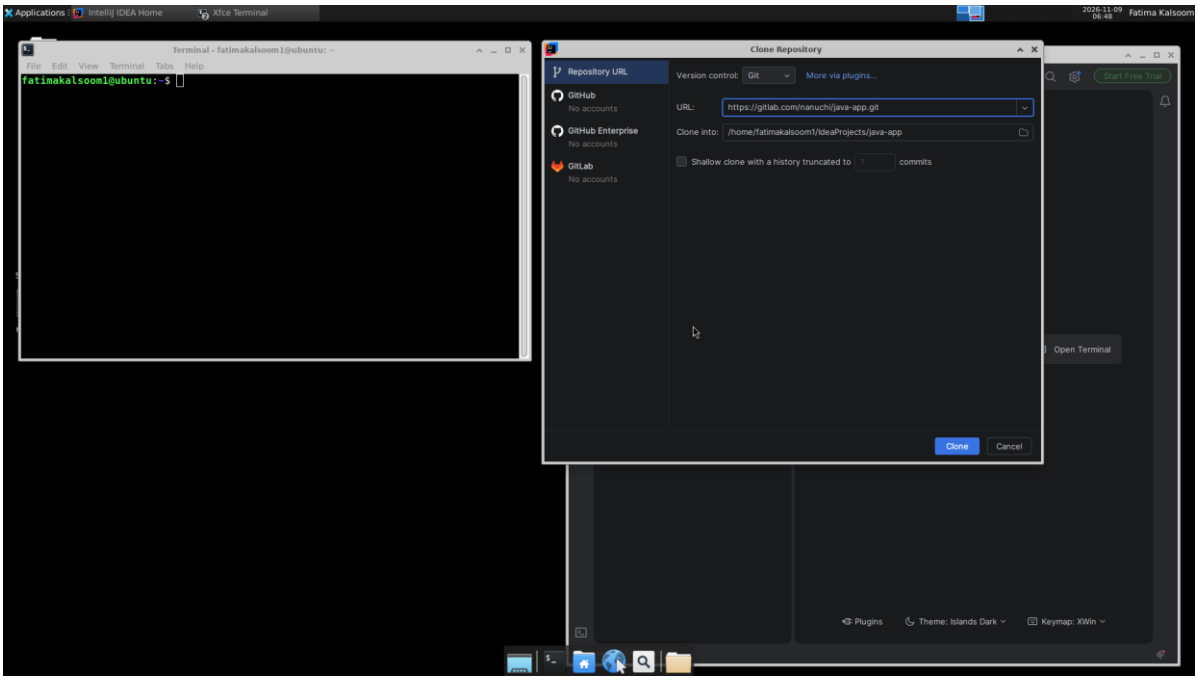


Running the Maven-built JAR with java -jar; the Spring Boot app starts on port 8080.

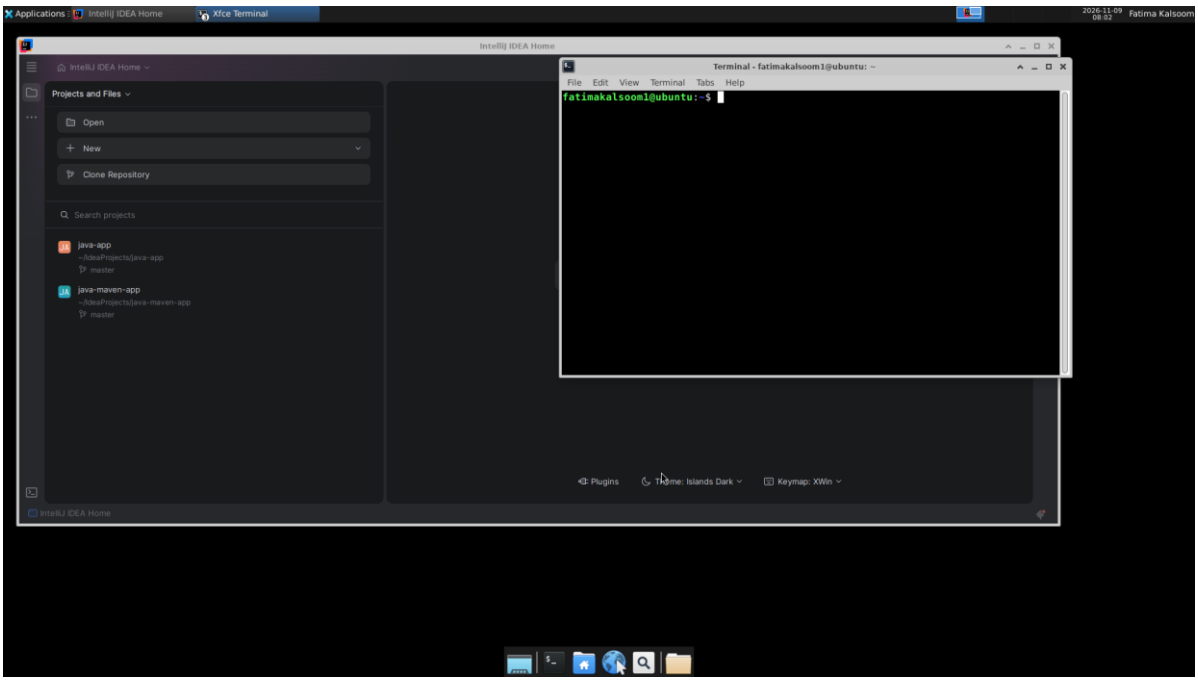


The running application stopped and the terminal returned to the prompt.

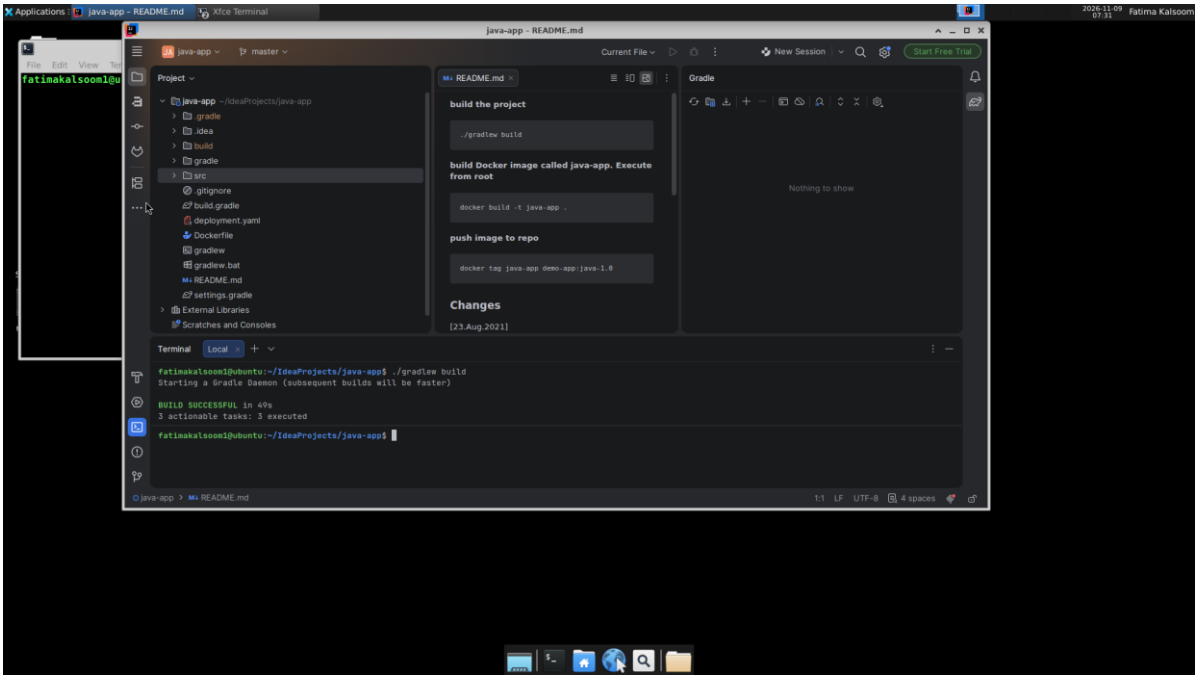
Task 2: Build a Java Application with Gradle



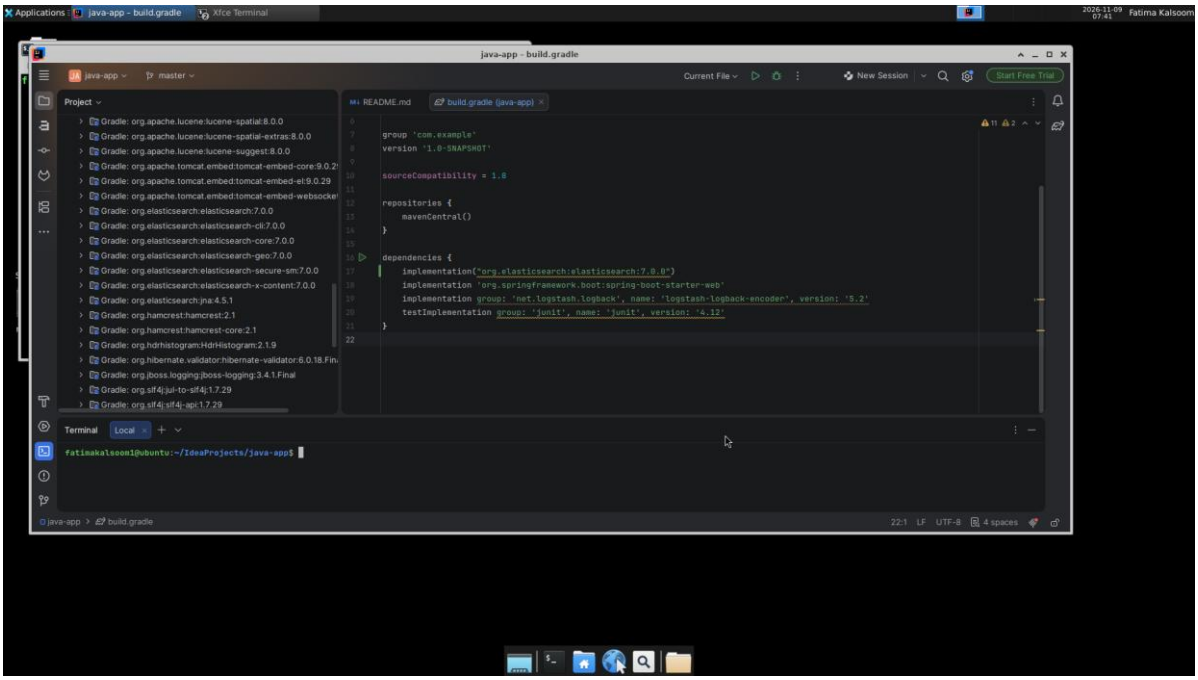
Cloning the java-app repository in IntelliJ IDEA.



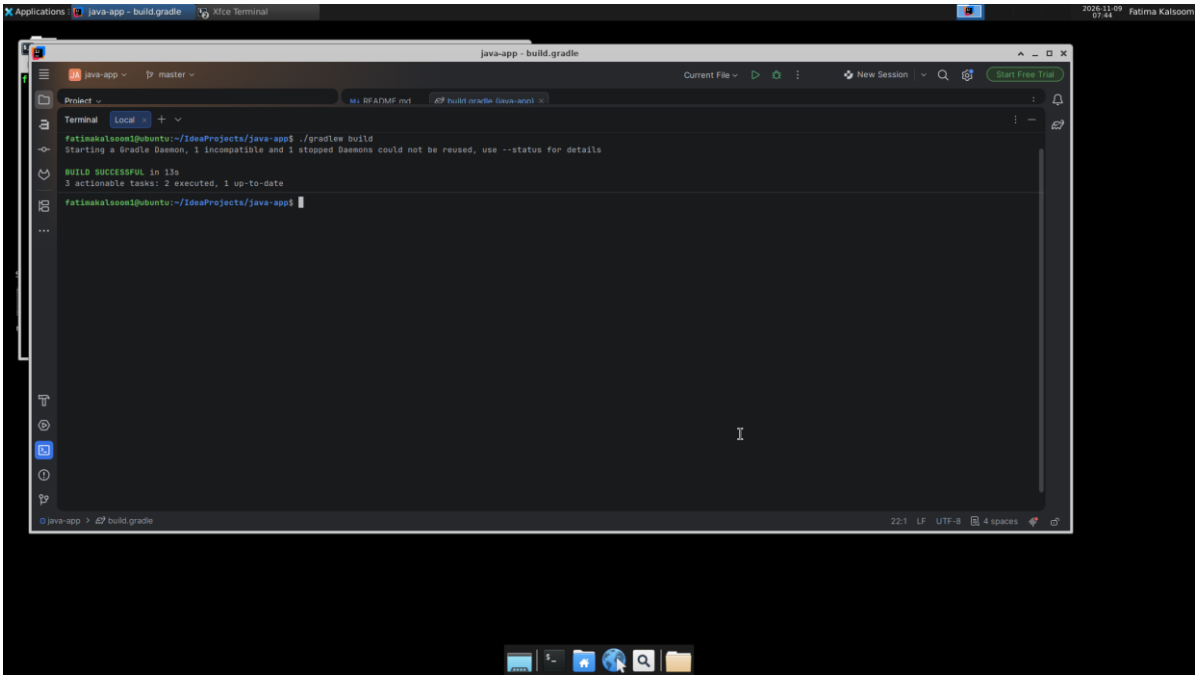
The java-app project listed on the IntelliJ IDEA welcome screen.



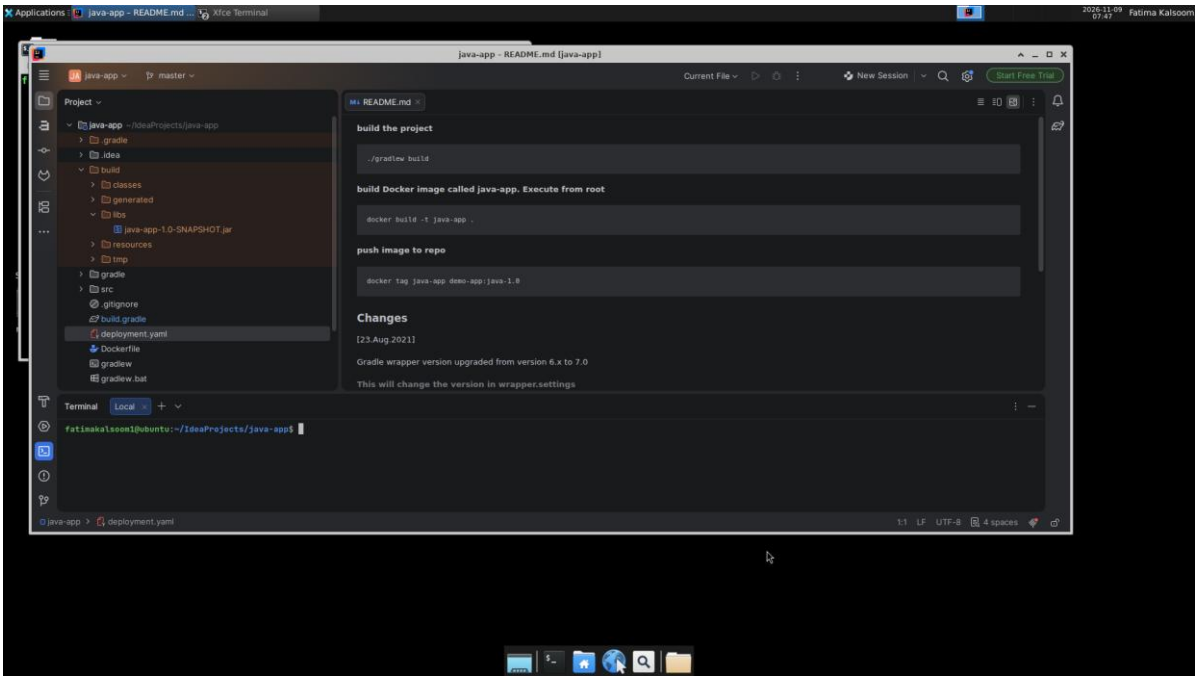
Building the project with ./gradlew build (BUILD SUCCESSFUL).



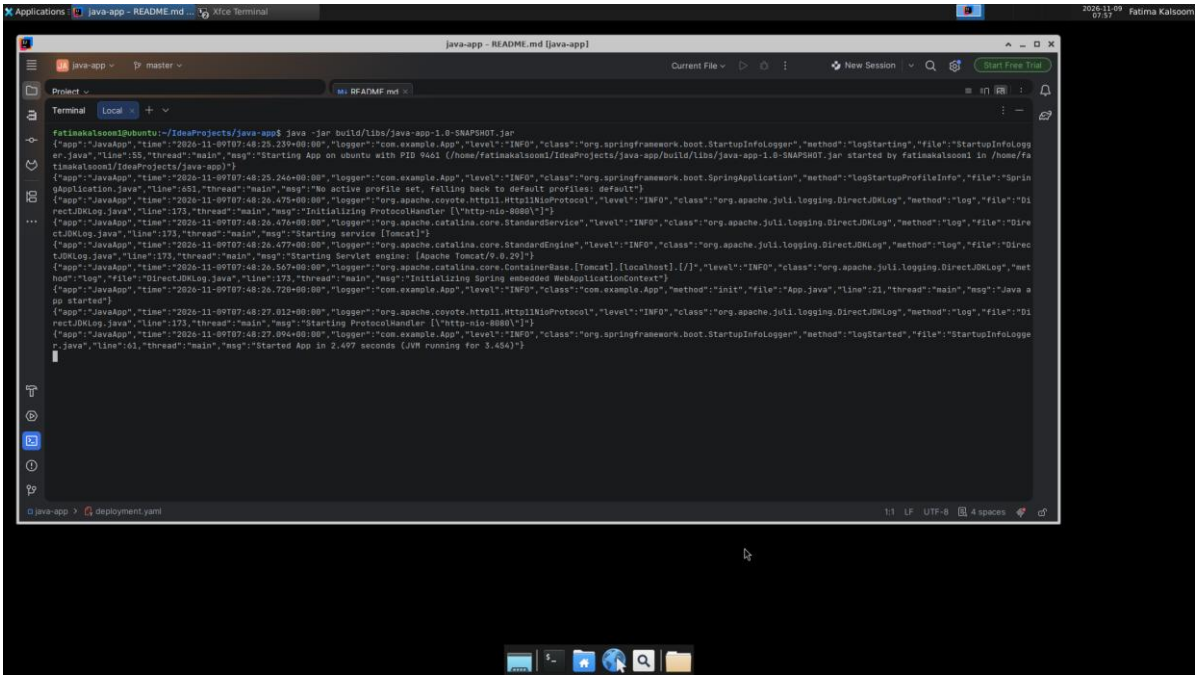
Elasticsearch dependency added in build.gradle and resolved in External Libraries.



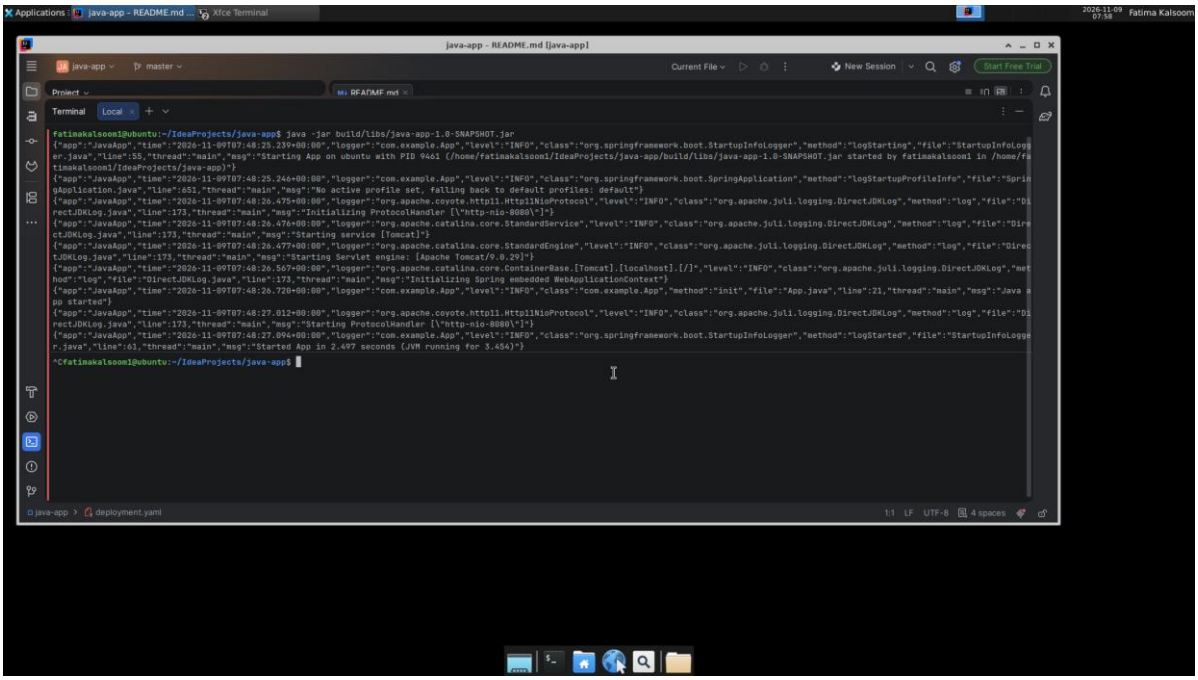
Re-building the project with ./gradlew build after the dependency change.



The generated JAR file inside the build/libs folder.

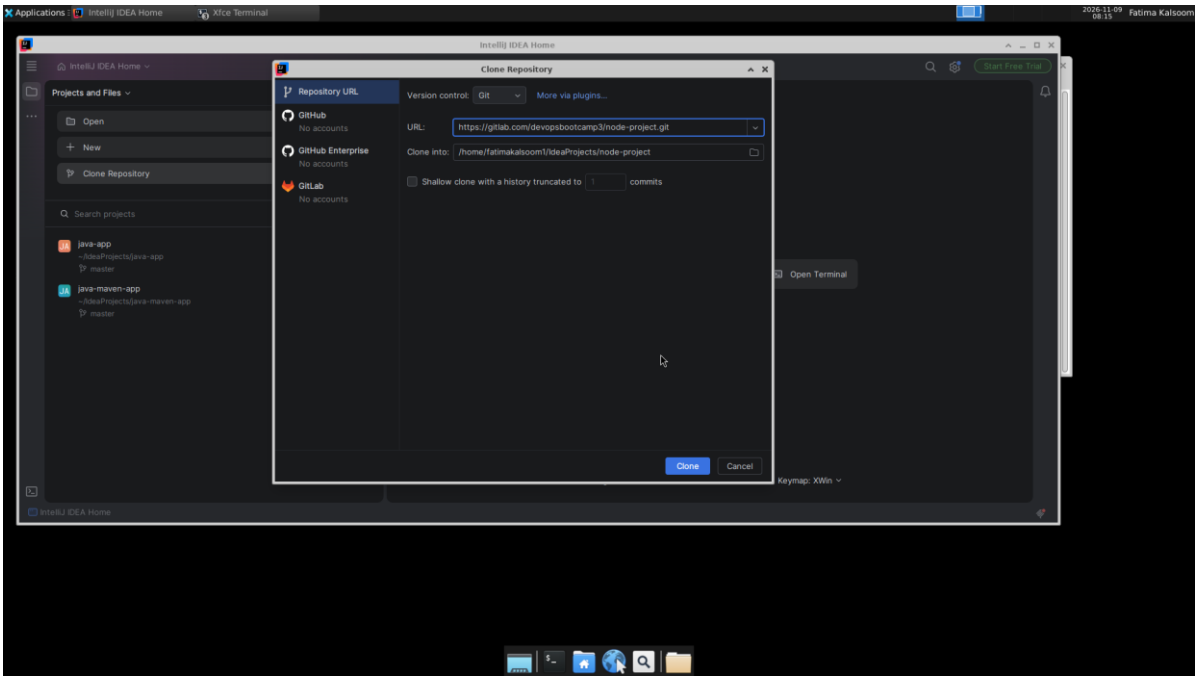


Running the Gradle-built JAR with java -jar; the application starts successfully.

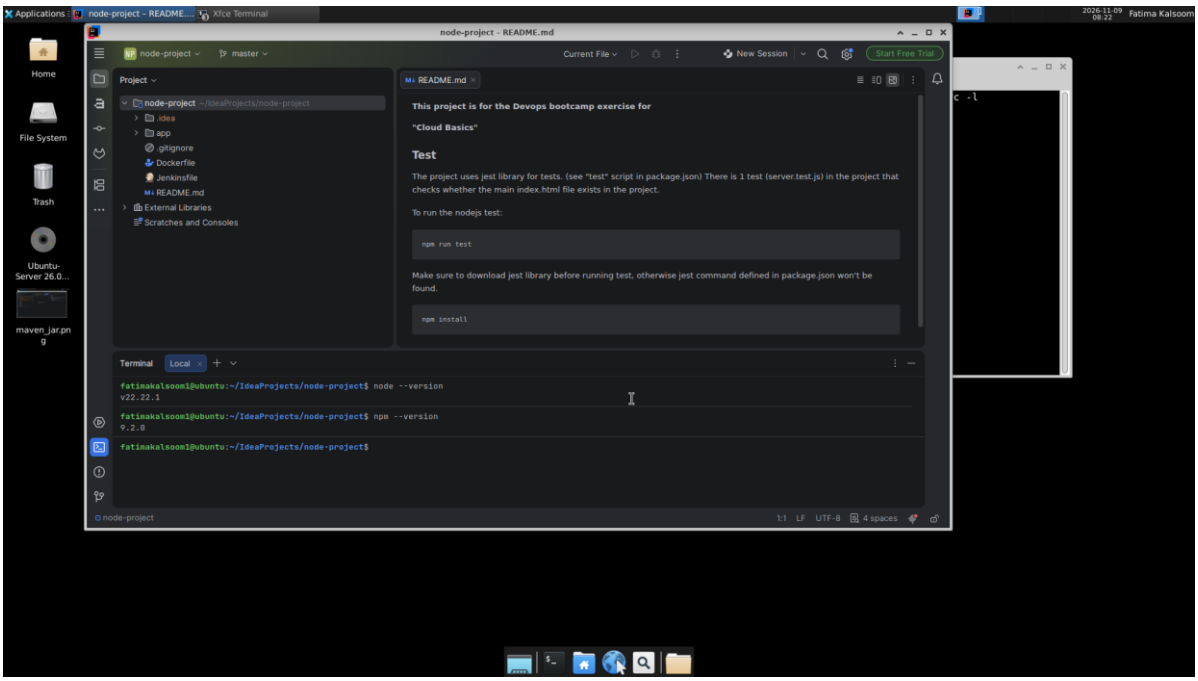


The running application stopped and the terminal returned to the prompt.

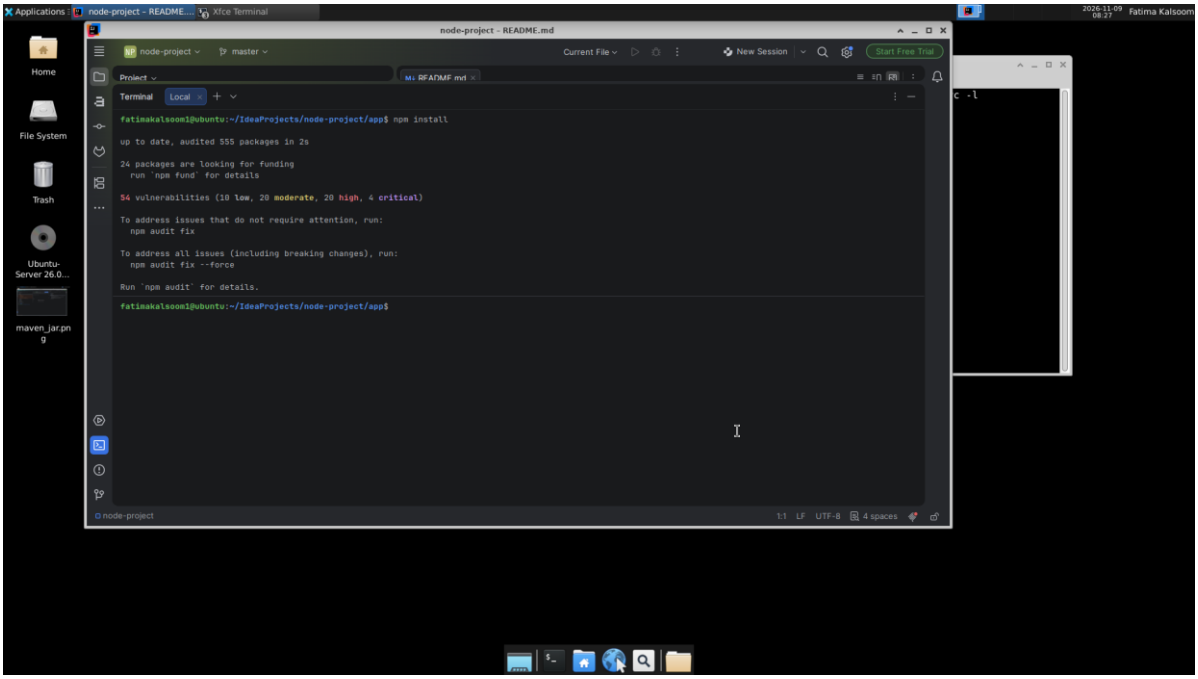
Task 3: Build a Node.js Application with npm



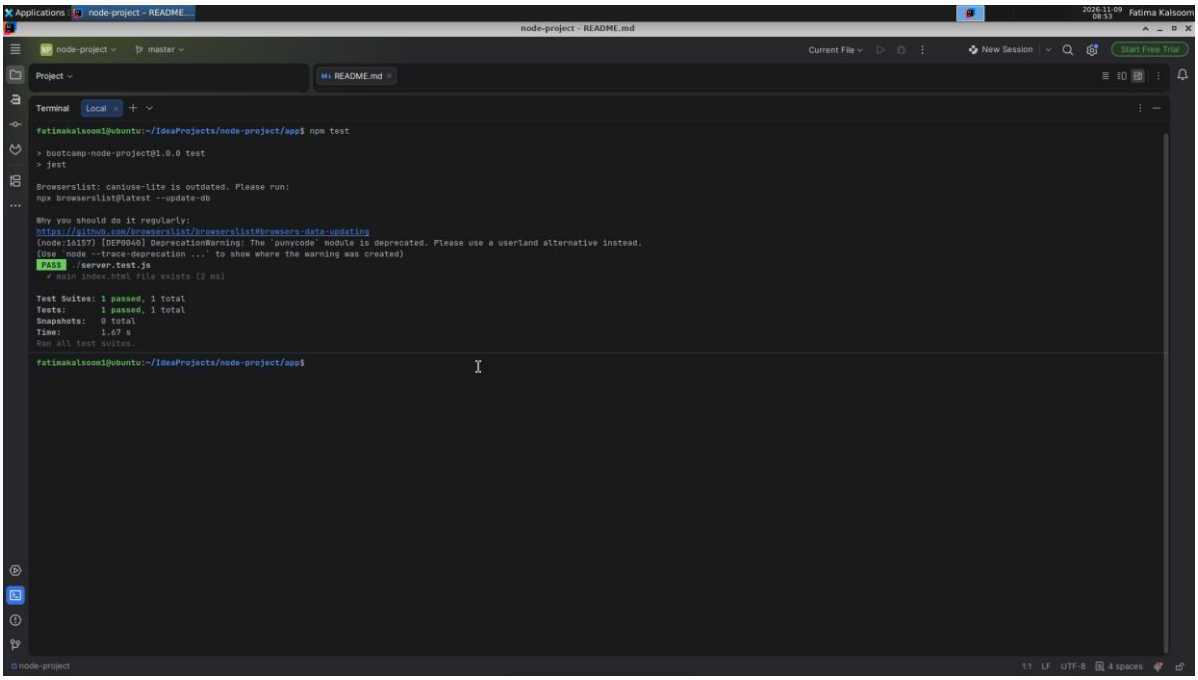
Cloning the node-project repository in IntelliJ IDEA.



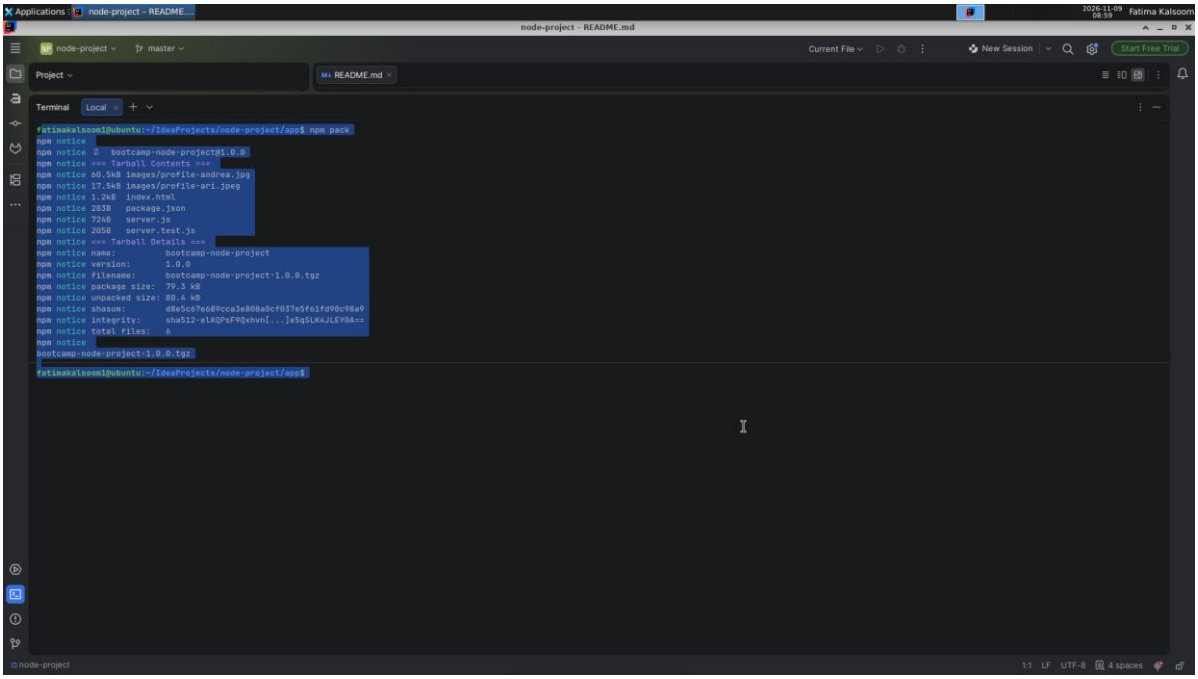
Checking the installed Node.js and npm versions.



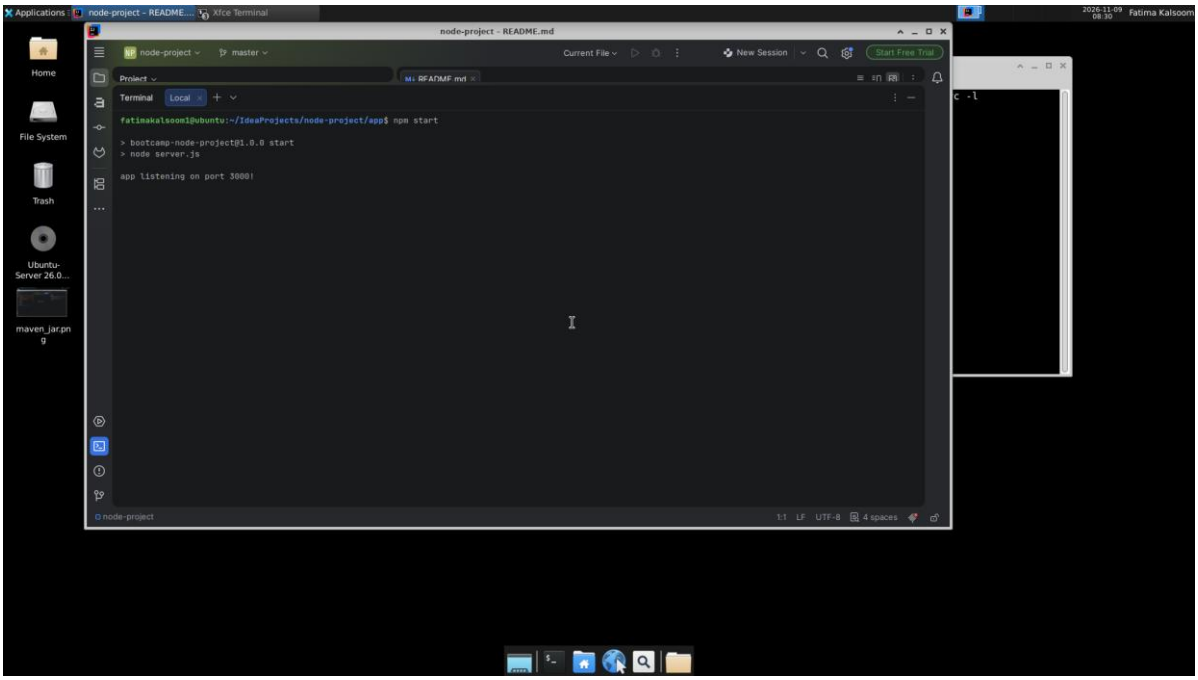
Installing the project dependencies with npm install.



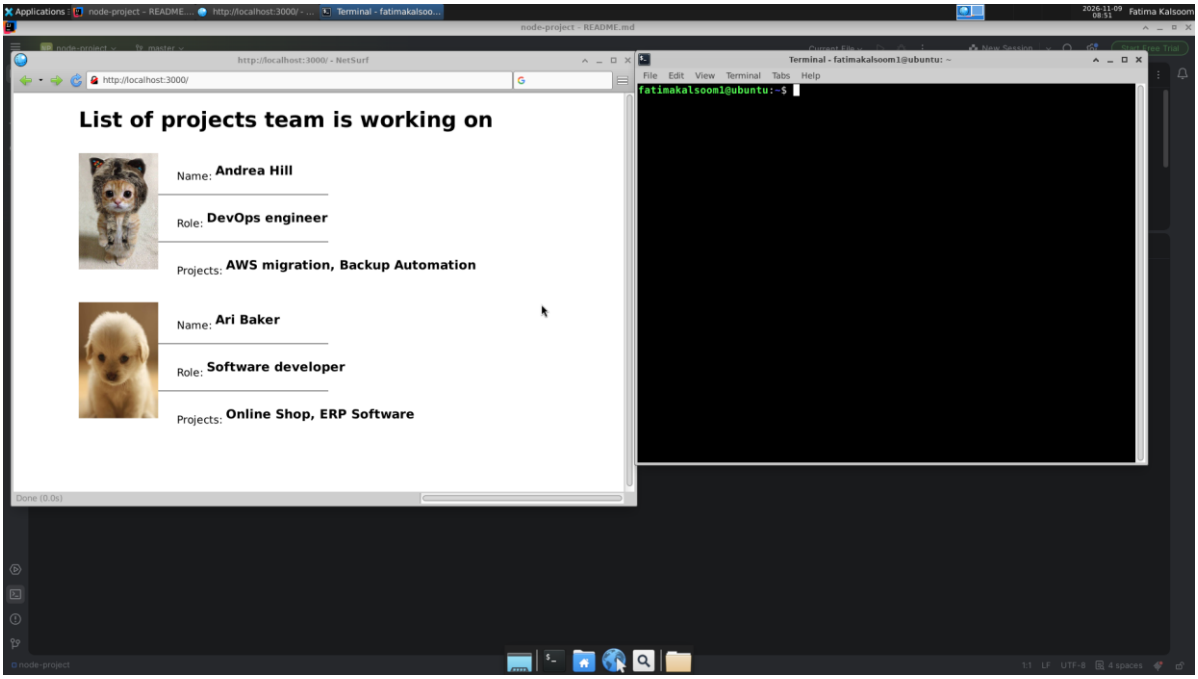
Running the tests with npm test (all tests passed).



Creating the package archive with npm pack.

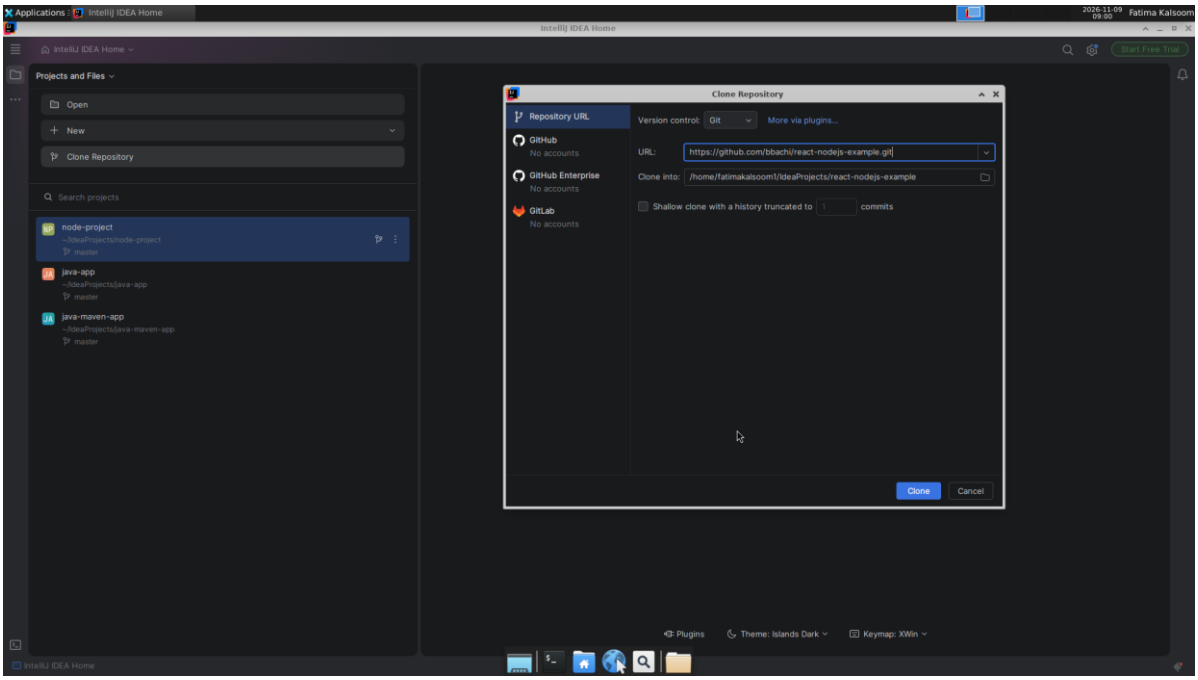


Starting the Node.js server with npm start; the app listens on port 3000.

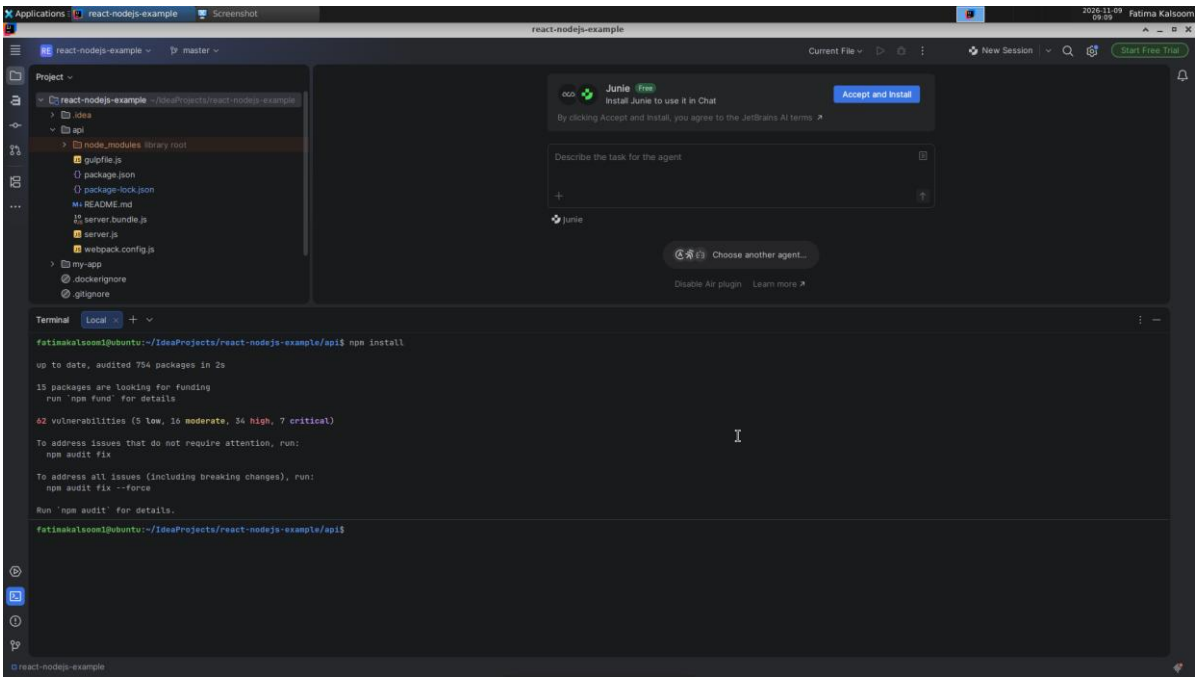


The application opened in the browser at localhost:3000.

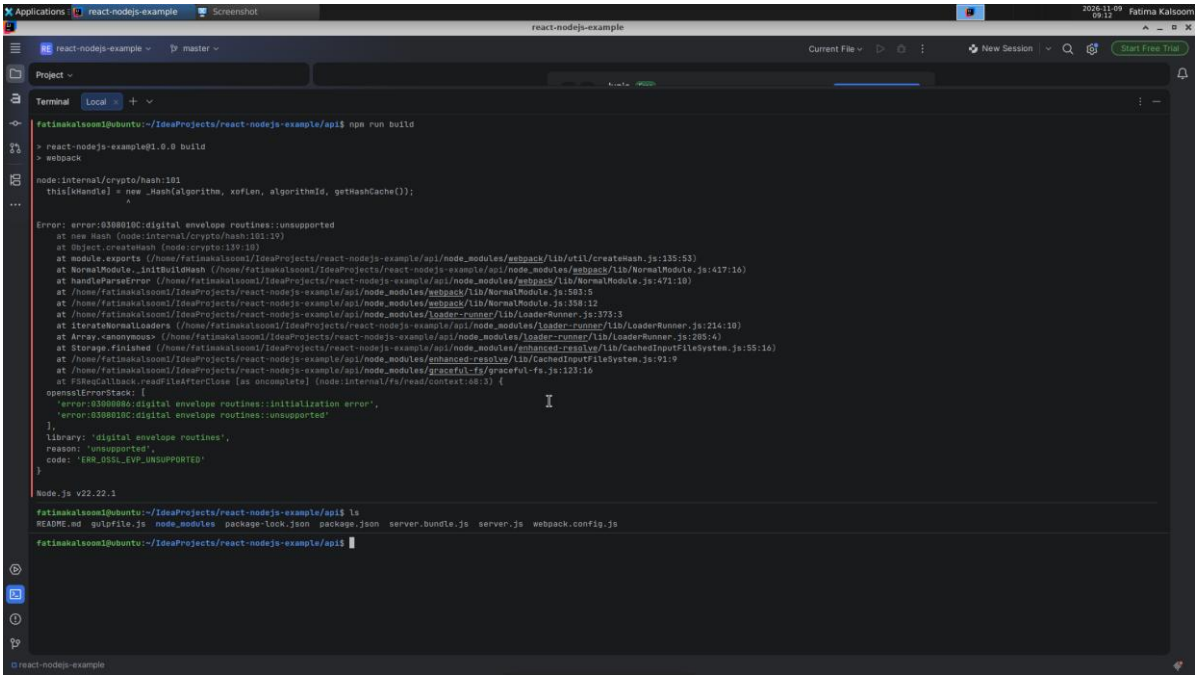
Task 4: Build a React and Node.js Application



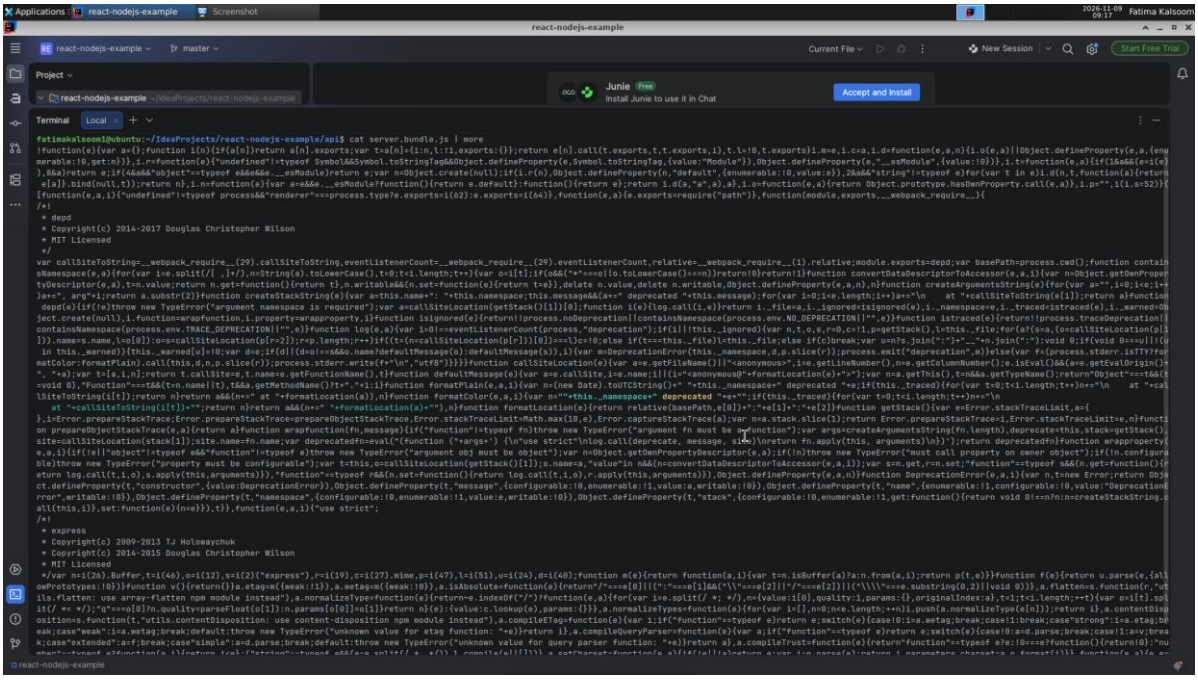
Cloning the react-nodejs-example repository in IntelliJ IDEA.



Installing the api folder dependencies with npm install.

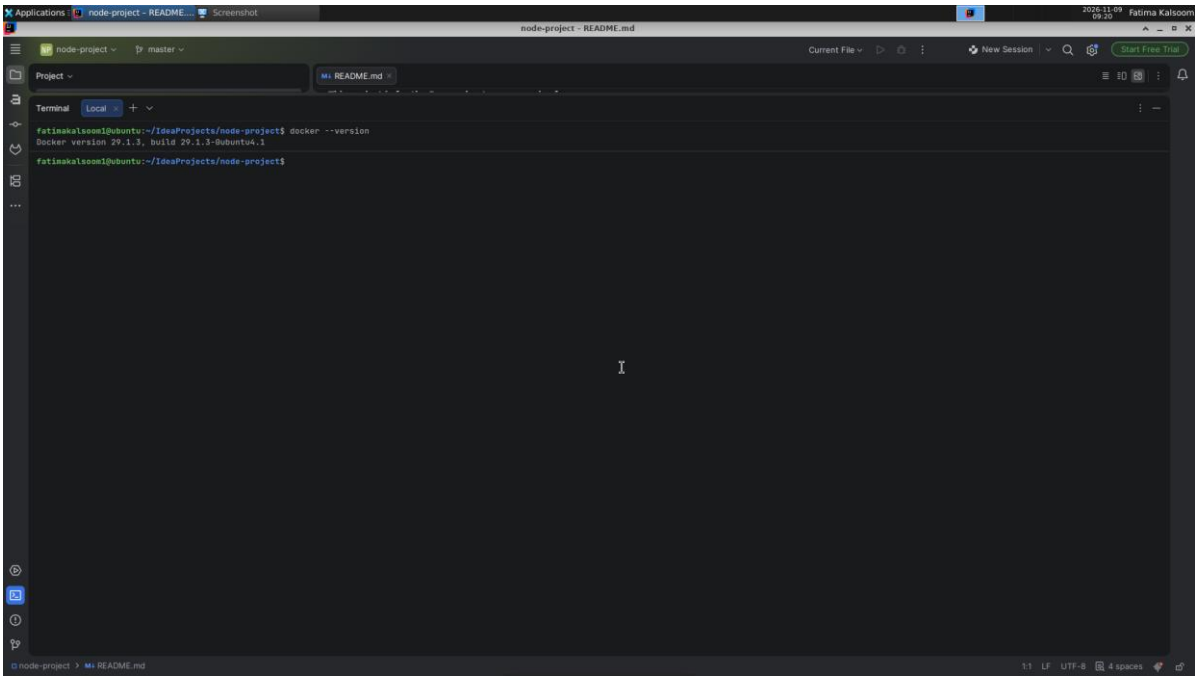


Building the bundle with npm run build.

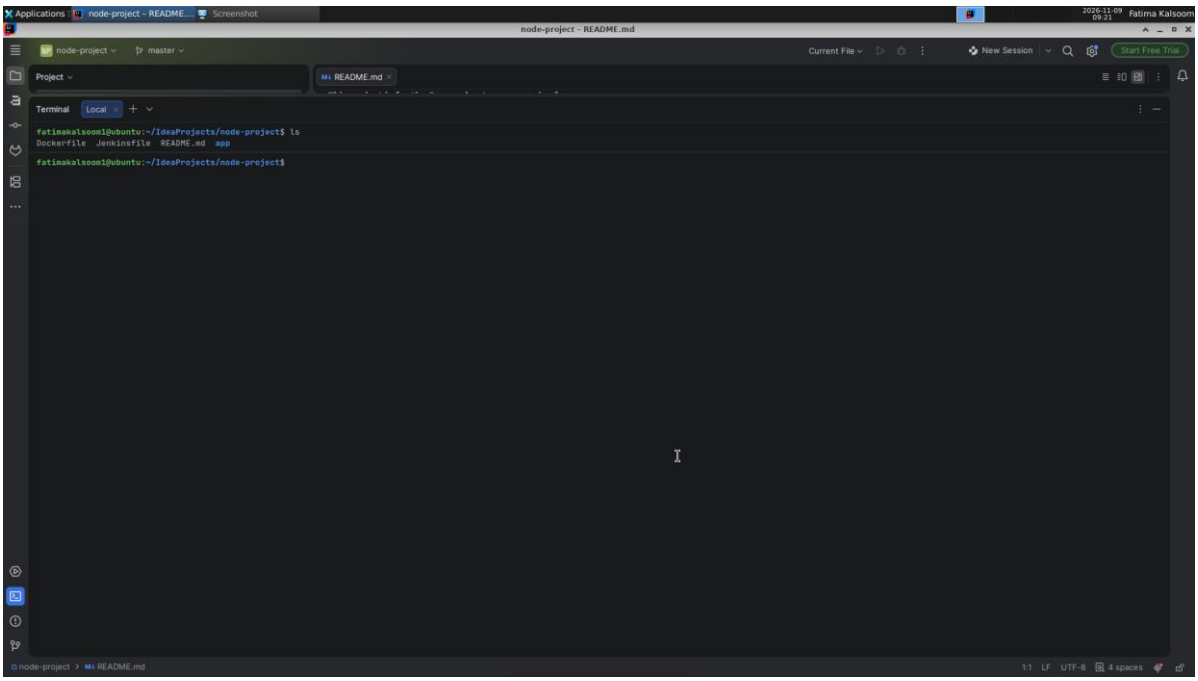


The generated server.bundle.js file output in the terminal.

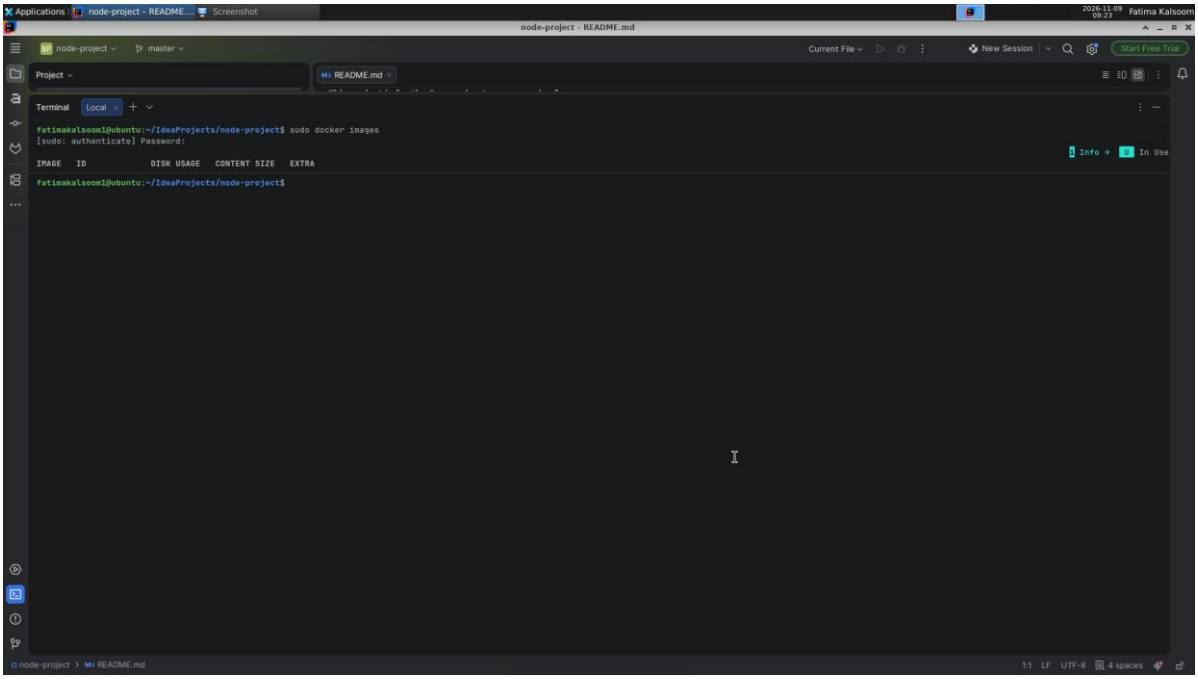
Task 5: Docker Image for the Node.js Application



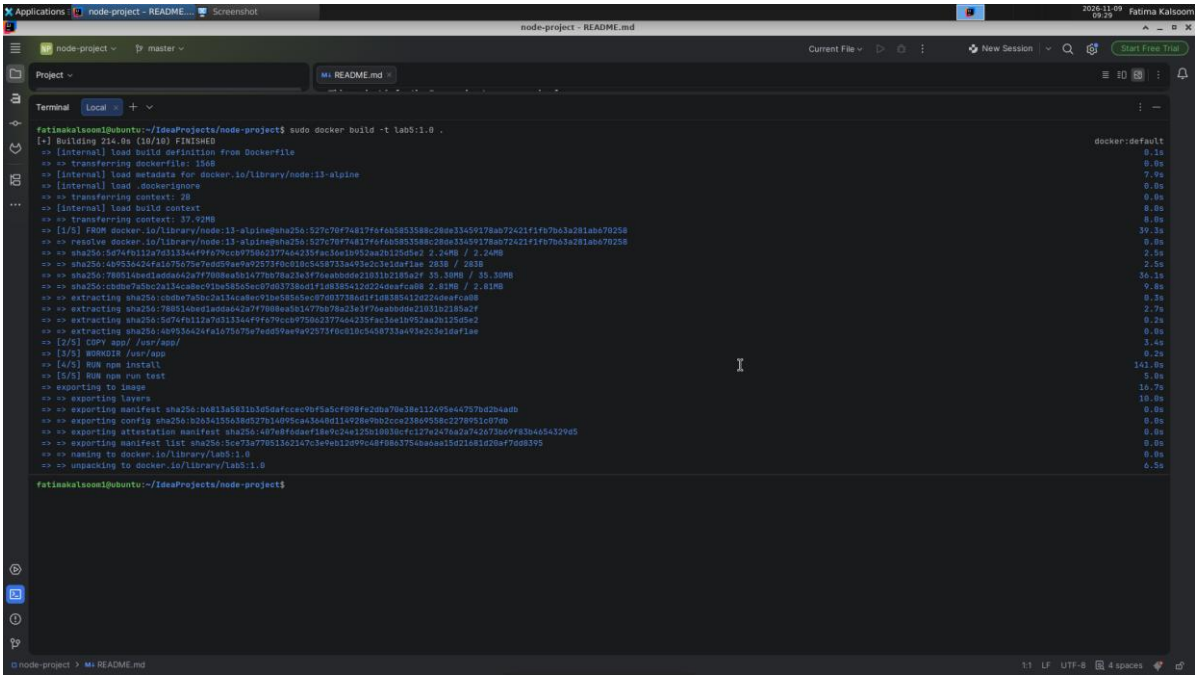
Verifying the Docker installation with docker --version.



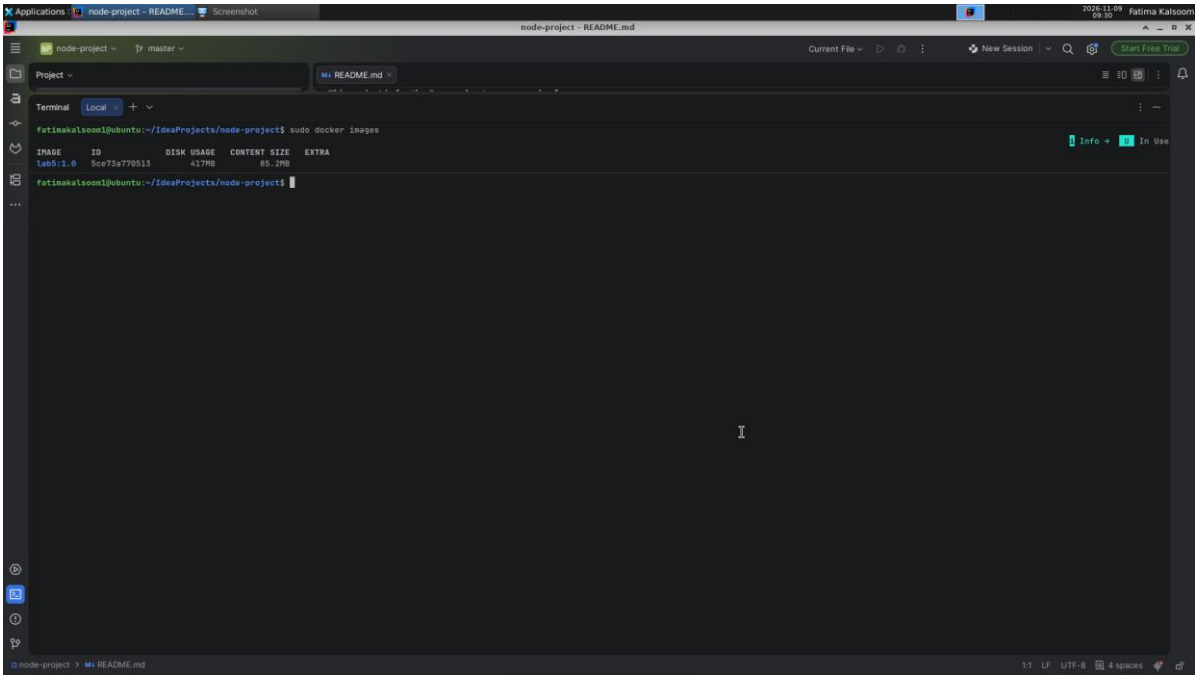
The Dockerfile present in the node-project folder.



Listing Docker images with docker images before the build (no images yet).

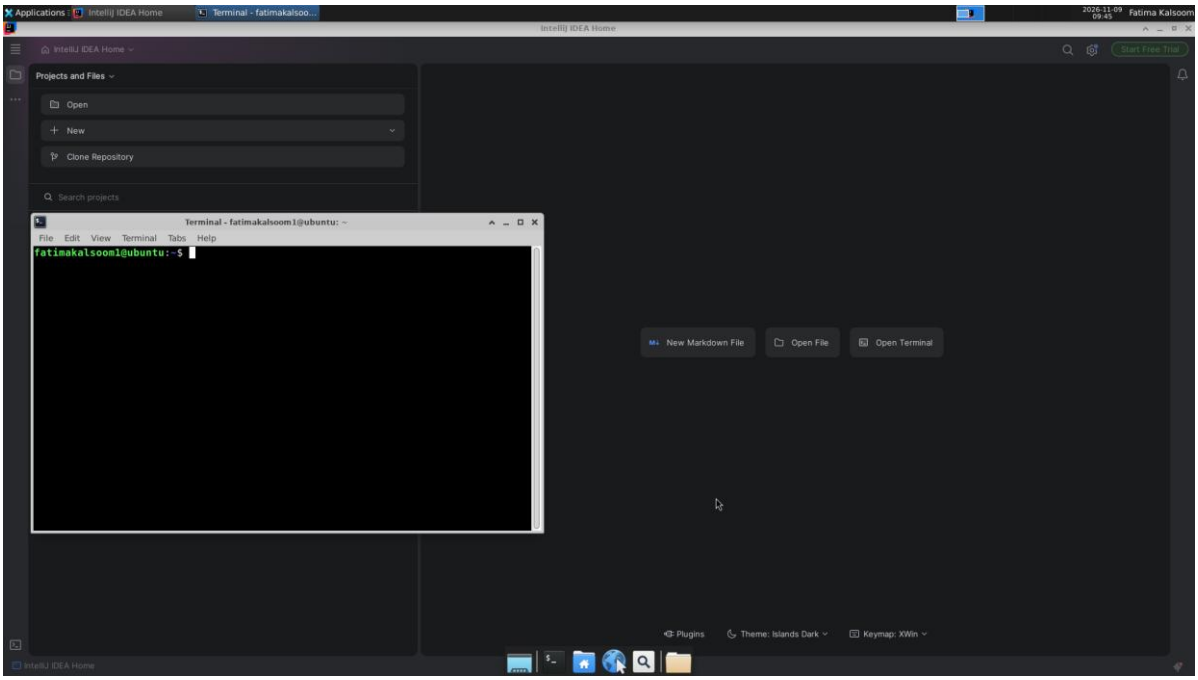


Building the Docker image with docker build.

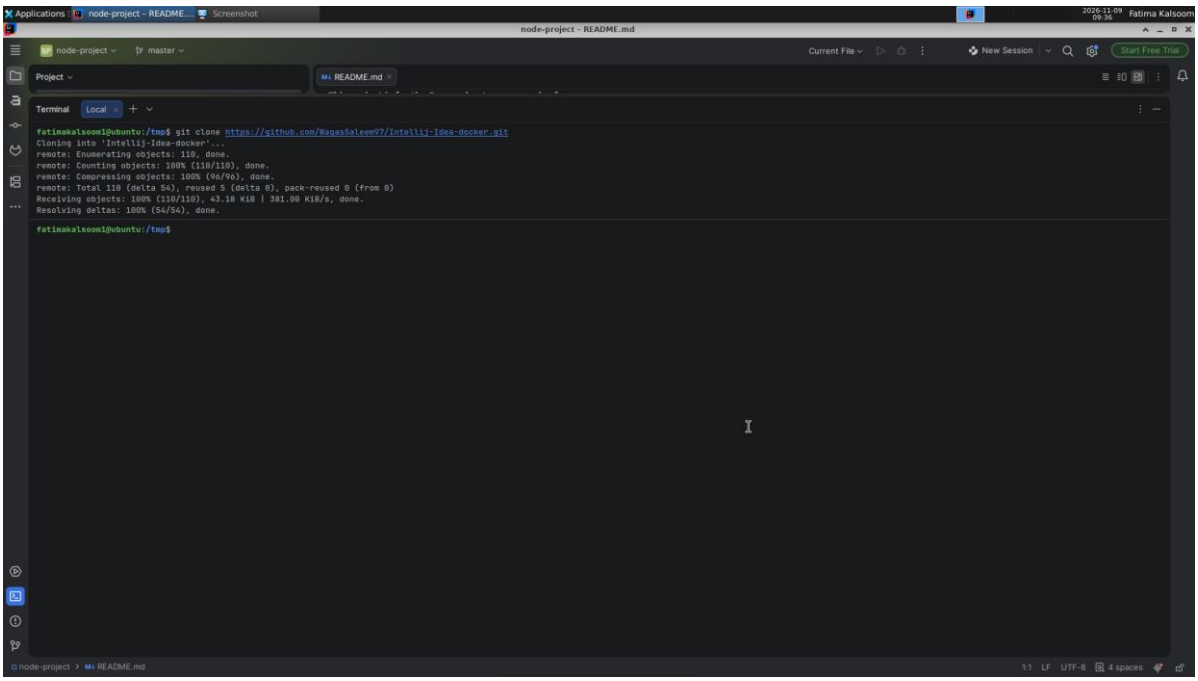


Listing Docker images after the build; the new image appears in the list.

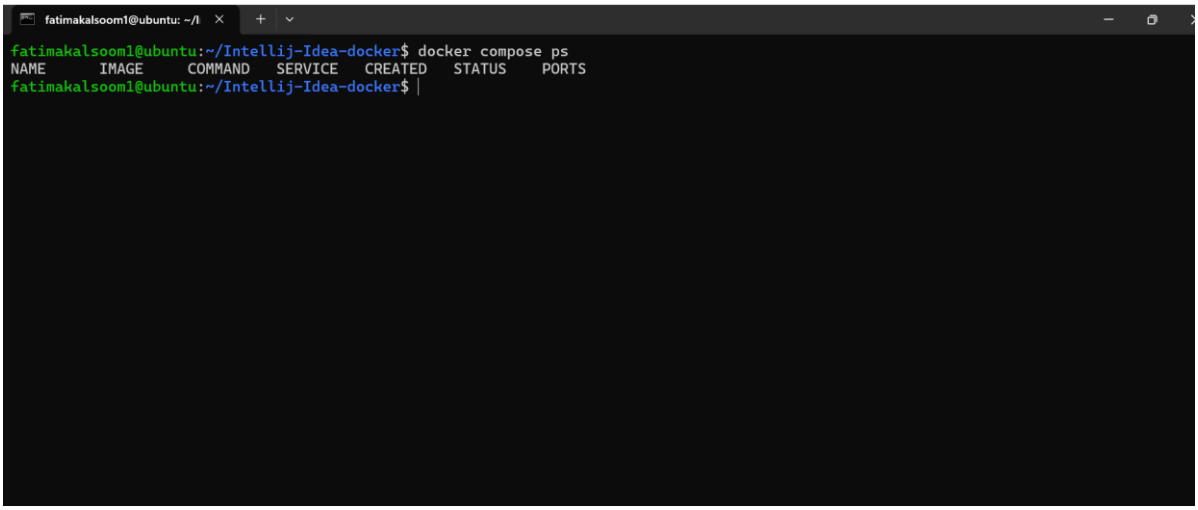
Task 6: IntelliJ IDEA in Docker



IntelliJ IDEA welcome screen with a terminal window open.



Cloning the Intellij-Idea-docker repository using git clone.



Checking the Docker Compose services with docker compose ps.